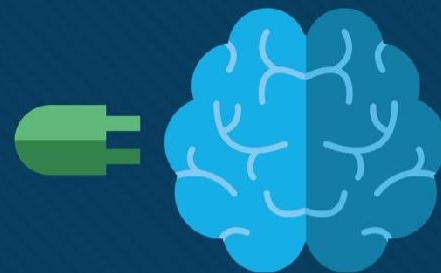




Chapter 4: Data Link Layer (OSI Model)

Introduction to Networks v7.02
(ITN)



Chapter 4 - Sections & Objectives

- 4.1 Data Link Layer Protocols
 - Explain the role of the data link layer in supporting communications across data networks.
 - Describe the purpose and function of the data link layer in preparing communication for transmission on specific media.
- 4.2 Media Access Control
 - Compare media access control techniques and logical topologies used in networks.
 - Compare the functions of logical topologies and physical topologies.
 - Describe the basic characteristics of media access control methods on WAN topologies.
 - Describe the basic characteristics of media access control methods on LAN topologies.
 - Describe the characteristics and functions of the data link frame.

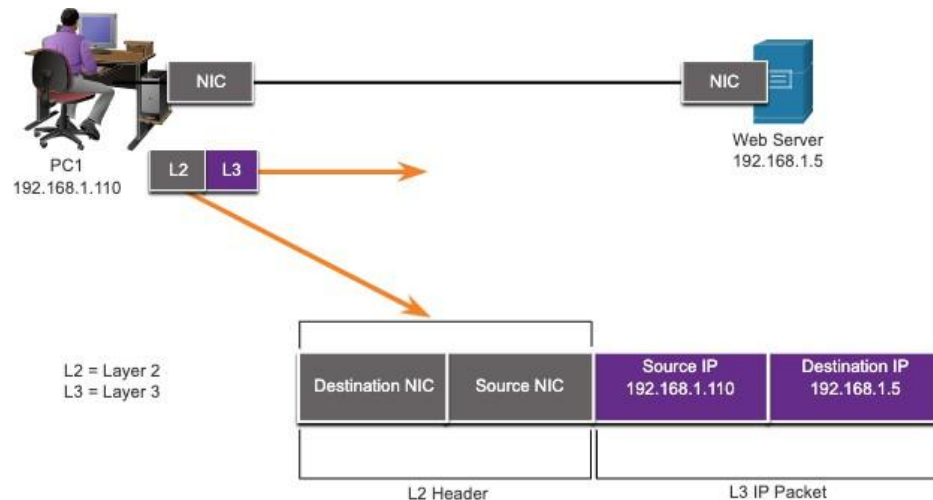
4.1 Data Link Layer Protocols

4.1.1 Purpose of the Data Link Layer

Purpose of the Data Link Layer

The Data Link Layer

- The Data Link layer is responsible for **communications between end-device network interface cards**.
- It allows upper layer protocols to access the physical layer media and **encapsulates Layer 3 packets (IPv4 and IPv6) into Layer 2 Frames**.
- It also performs **error detection and rejects corrupts frames**.

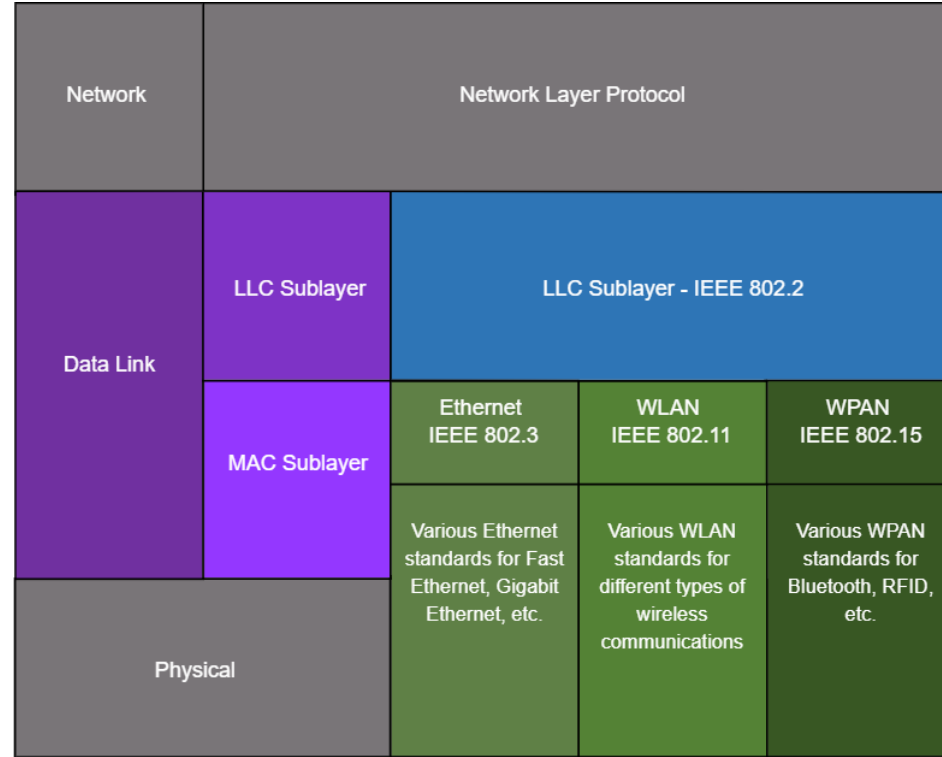


IEEE 802 LAN/MAN Data Link Sublayers

IEEE 802 LAN/MAN standards are specific to the type of network (Ethernet, WLAN, WPAN, etc).

The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC)**.

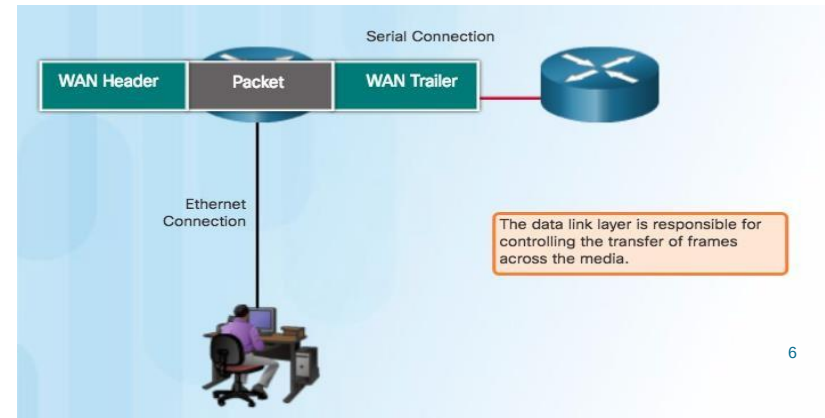
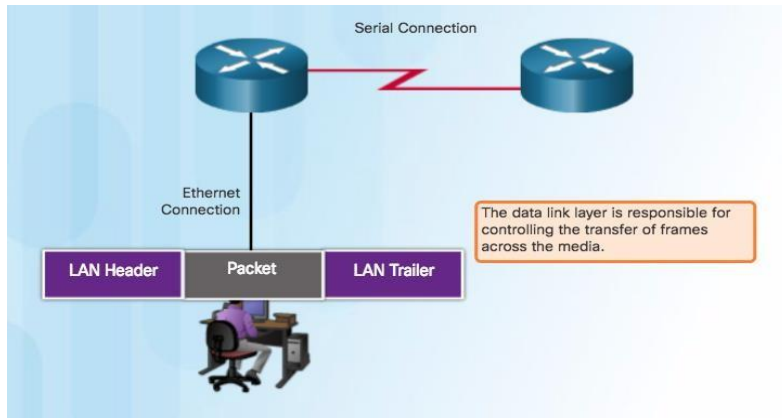
- The LLC sublayer **communicates between the networking software at the upper layers and the device hardware at the lower layers.**
- The MAC sublayer is **responsible for data encapsulation and media access control.**



Purpose of the Data Link Layer

Providing Access to Media

- Packets exchanged between nodes may experience numerous data link layers and media transitions.
- At **each hop** along the path, a **router performs four basic Layer 2 functions**:
 - Accepts a frame from the network medium.
 - De-encapsulates the frame to expose the encapsulated packet.
 - Re-encapsulates the packet into a new frame.
 - Forwards the new frame on the medium of the next network segment.



Purpose of the Data Link Layer

Data Link Layer Standards

Data link layer protocols are defined by engineering organizations:

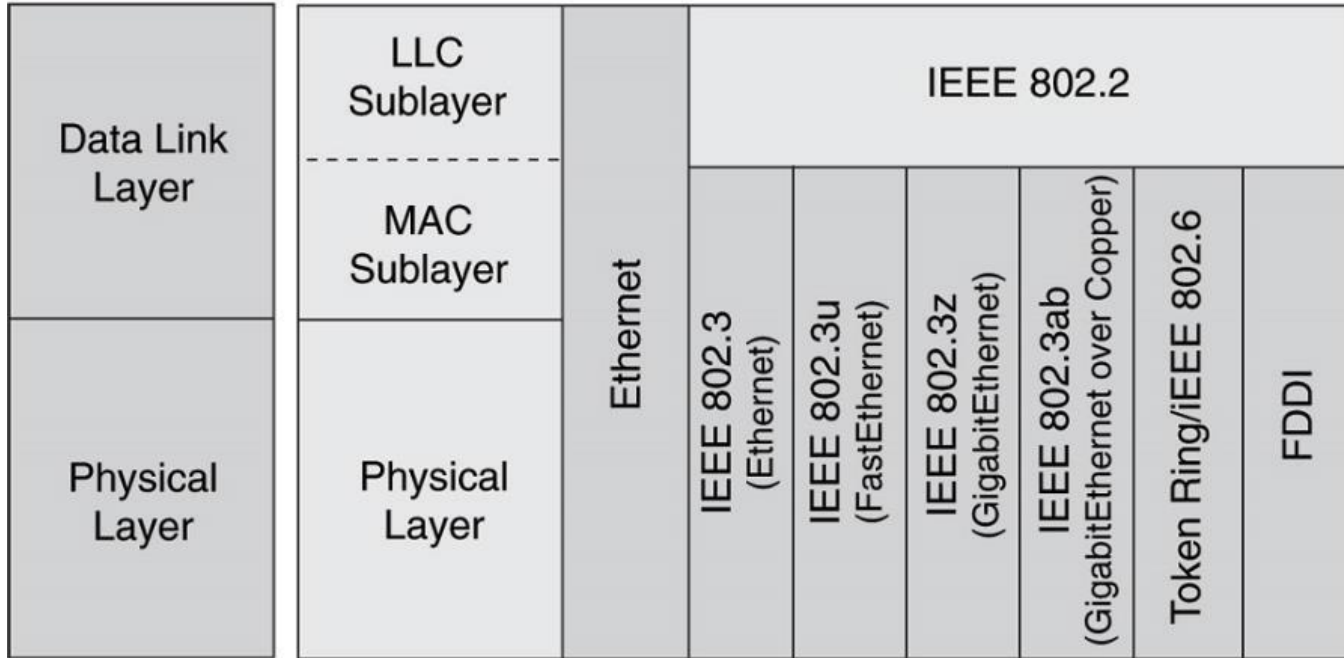
- Institute for Electrical and Electronic Engineers (IEEE).
- International Telecommunications Union (ITU).
- International Organizations for Standardization (ISO).
- American National Standards Institute (ANSI).



Data Link Layer Standards

Standard organization	Networking Standards
IEEE	• 802.2: Logical Link Control (LLC) • 802.3: Ethernet • 802.4: Token bus • 802.5: Token passing • 802.11: Wireless LAN (WLAN) & Mesh (Wi-Fi certification) • 802.15: Bluetooth • 802.16: WiMax
ITU-T	• G.992: ADSL • G.8100 - G.8199: MPLS over Transport aspects • Q.921: ISDN • Q.922: Frame Relay
ISO	• HDLC (High Level Data Link Control) • ISO 9314: FDDI Media Access Control (MAC)
ANSI	• X3T9.5 and X3T12: Fiber Distributed Data Interface (FDDI)

Layer 2 Standards



OSI Layers

LAN Specification

4.2 Topologies and Media Access Control

Topologies

Physical and Logical Topologies

The topology of a network is the arrangement and relationship of the network devices and the interconnections between them.

There are two types of topologies used when describing networks:

- **Physical topology** – shows physical connections and how devices are interconnected.
- **Logical topology** – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

Topologies

WAN Topologies

There are three common physical WAN topologies:

- **Point-to-point** – the simplest and most common WAN topology. Consists of a permanent link between two endpoints.
- **Hub and spoke** – similar to a star topology where a central site interconnects branch sites through point-to-point links.
- **Mesh** – provides high availability but requires every end system to be connected to every other end system.

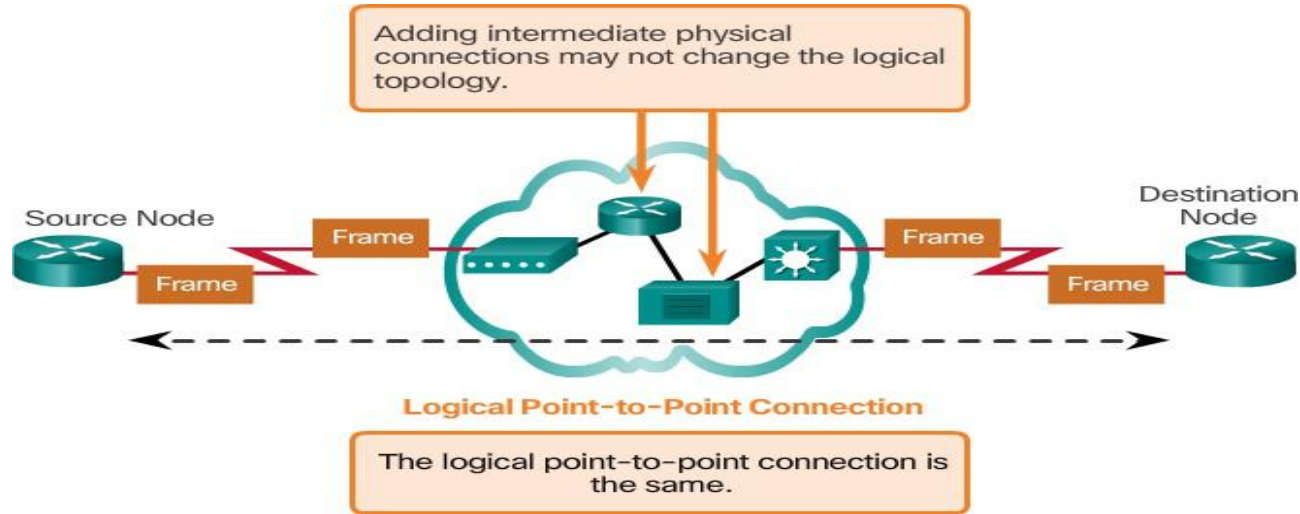
Topologies

Point-to-Point WAN Topology

- Physical point-to-point topologies directly connect two nodes.
- The nodes may not share the media with other hosts.
- Because **all frames on the media can only travel to or from the two nodes**, Point-to-Point WAN protocols can be very simple.



Logical Point-to-Point Topology



- The media access method used by the data link protocol is determined by the logical point-to-point topology, not the physical topology.
- This means that the logical point-to-point connection between two nodes may not necessarily be between two physical nodes at each end of a single physical link.

Topologies

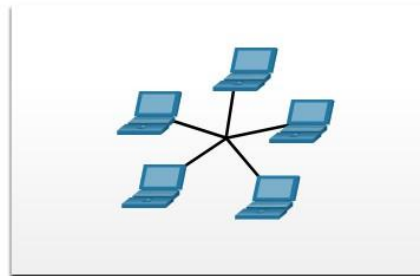
LAN Topologies

End devices on LANs are typically interconnected using a star or extended star topology. Star and extended star topologies are easy to install, very scalable and easy to troubleshoot.

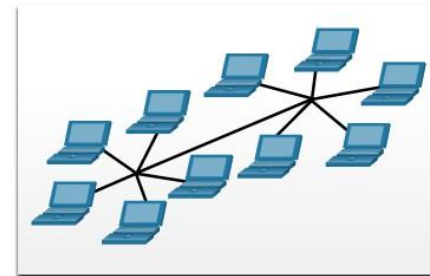
Early Ethernet and Legacy Token Ring technologies provide two additional topologies:

- **Bus** – All end systems chained together and terminated on each end.
- **Ring** – Each end system is connected to its respective neighbors to form a ring.

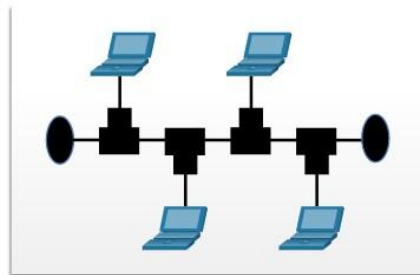
Physical Topologies



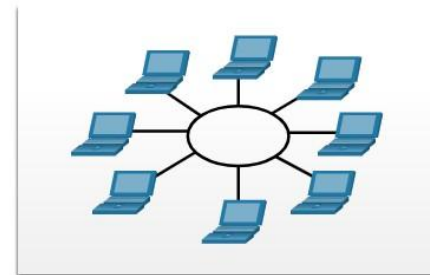
Star Topology



Extended Star Topology



Bus Topology

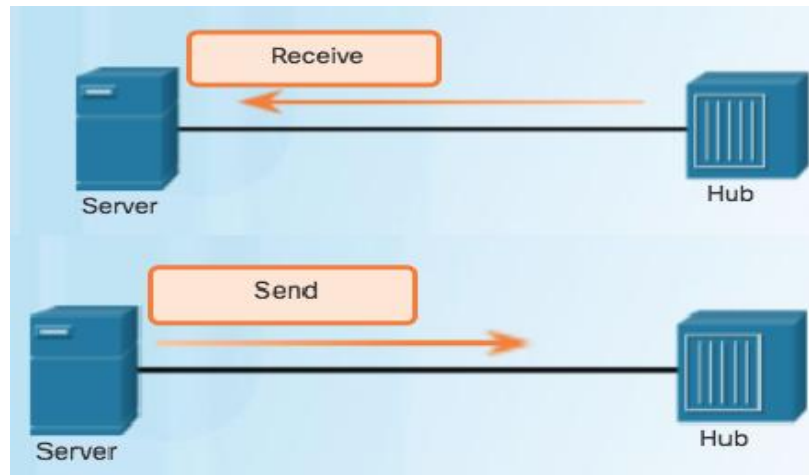


Ring Topology

Half and Full Duplex Communication

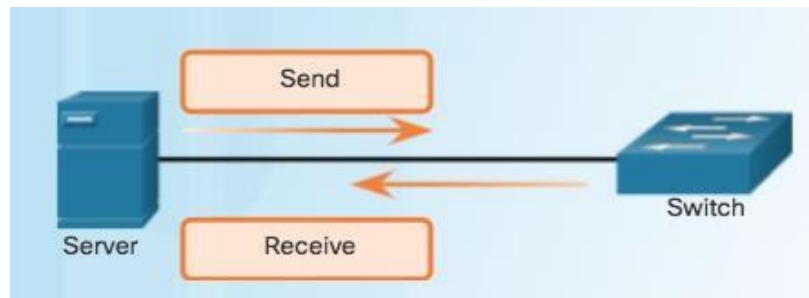
Half-duplex communication

- Only allows one device to send or receive at a time on a shared medium.
- Used on WLANs and legacy bus topologies with **Ethernet hubs**.



Full-duplex communication

- Allows both devices to simultaneously transmit and receive on a shared medium.
- **Ethernet switches** operate in full-duplex mode.



Contention-based access

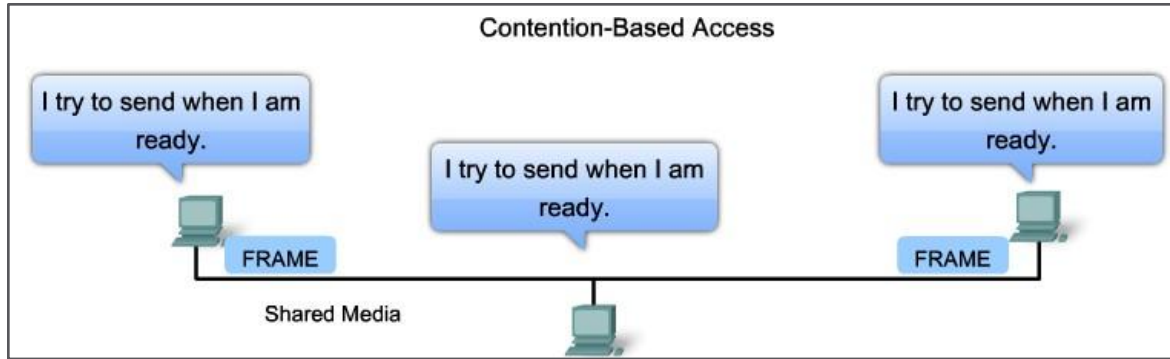
All nodes operating in half-duplex, competing for use of the medium. Examples are:

- Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
- Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

Controlled access

- Deterministic access where each node has its own time on the medium.
- Used on legacy networks such as Token Ring and ARCNET.

Contention-Based Access



Characteristics	Contention-Based Technologies
• Stations can transmit at any time • Collision exist • There are mechanisms to resolve contention for the media	• CSMA/CD for 802.3 Ethernet networks WITH HUB, • if you wrongly configure your switch from full duplex to become half duplex, then use contention based access method • CSMA/CA for 802.11 wireless networks

Topologies

Contention-Based Access – CSMA/CD

CSMA/CD

- Used by legacy Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CD collision detection process:

- Devices transmitting simultaneously will result in a signal collision on the shared media.
- Devices detect the collision.
- Devices wait a random period of time and retransmit data.

Topologies

Contention-Based Access – CSMA/CA

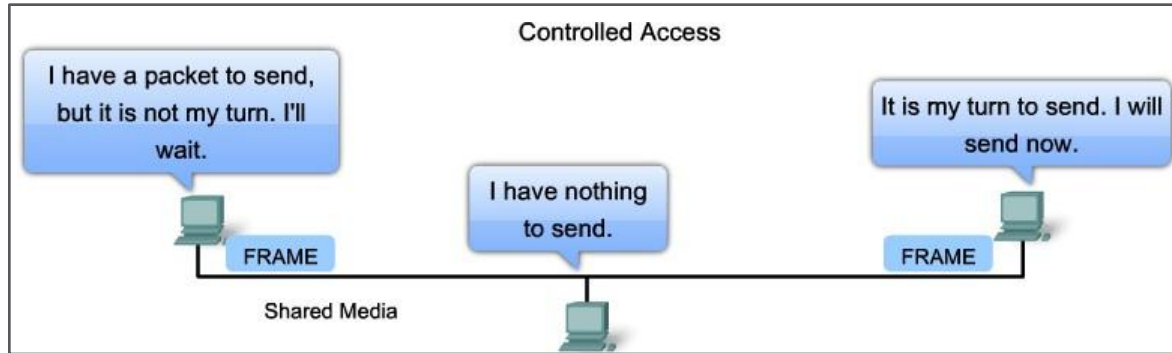
CSMA/CA

- Used by IEEE 802.11 WLANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CA collision avoidance process:

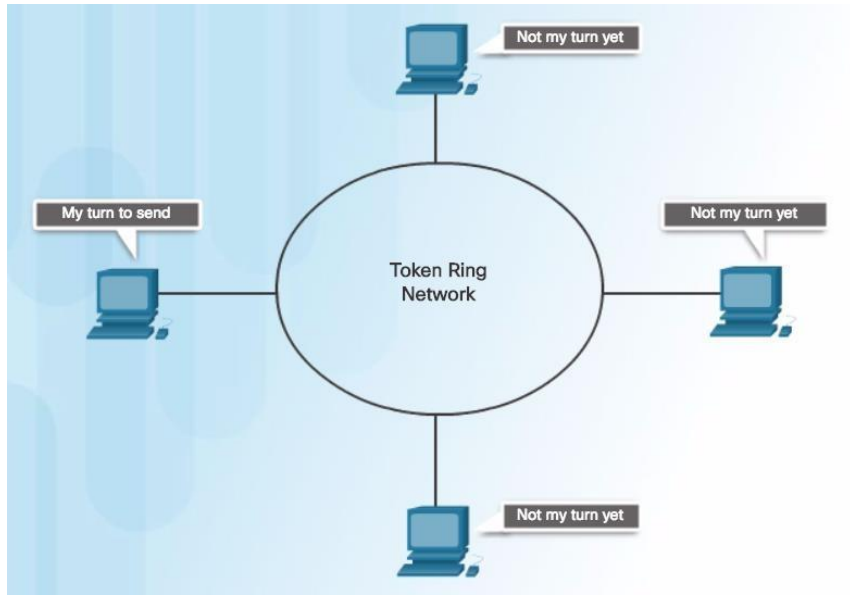
- When transmitting, devices also include the time duration needed for the transmission.
- Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

Controlled Access



Characteristics	Controlled Access Technologies
• Only one station can transmit at a time • Devices wishing to transmit must wait their turn • No collisions • May use a token passing method	• Token Ring (IEEE 802.5) • Fiber Distributed Data Interface (FDDI)

Media Access Control Methods (Cont.)



■ Controlled Access

- Each node has its own time to use the medium.
- Legacy Token Ring LANs are an example

6.3 Data Link Frame

Data Link Frame

The Frame

Data is encapsulated by the data link layer with a header and a trailer to form a frame.

A data link frame has three parts:

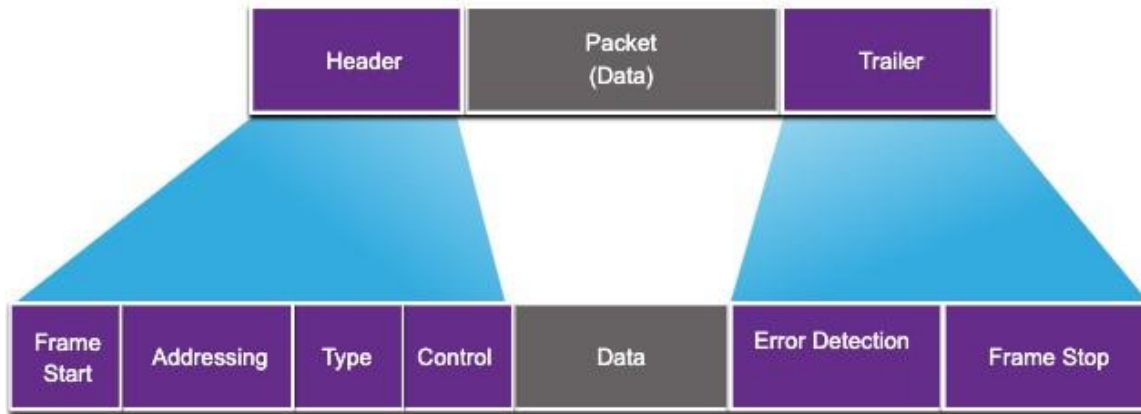
- Header
- Data
- Trailer

The fields of the header and trailer vary according to data link layer protocol.

The amount of control information carried with in the frame varies according to access control information and logical topology.

Data Link Frame

Frame Fields

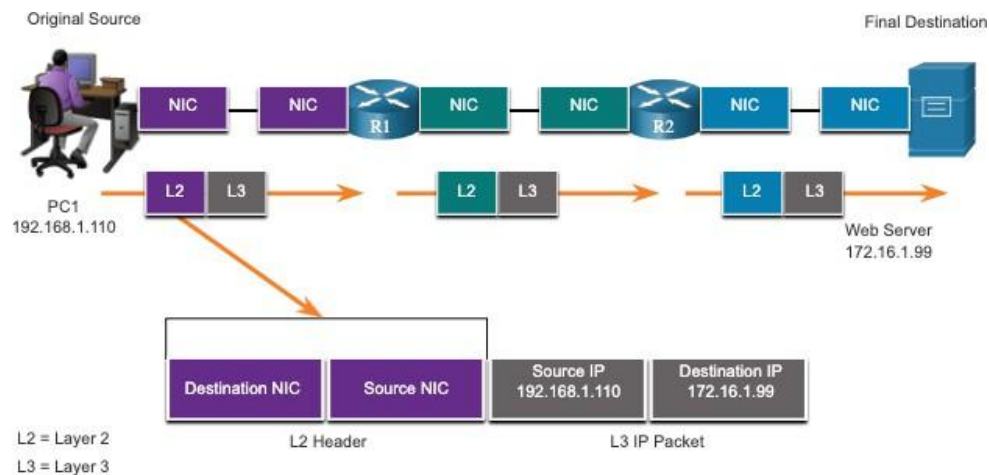


Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

Data Link Frame

Layer 2 Addresses

- Also referred to as a physical address.
- Contained in the frame header.
- Used only for local delivery of a frame on the link.
- Updated by each device that forwards the frame.



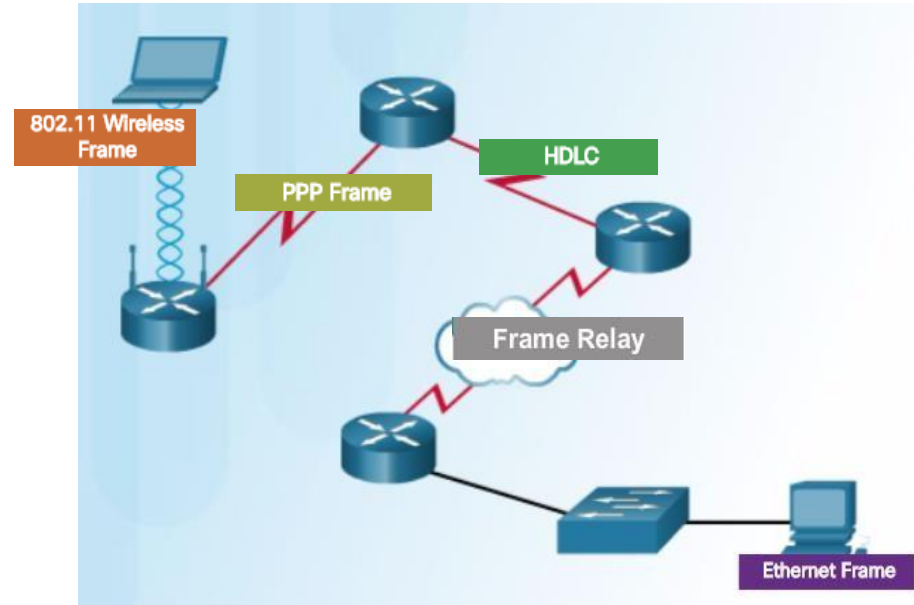
Data Link Frame

LAN and WAN Frames

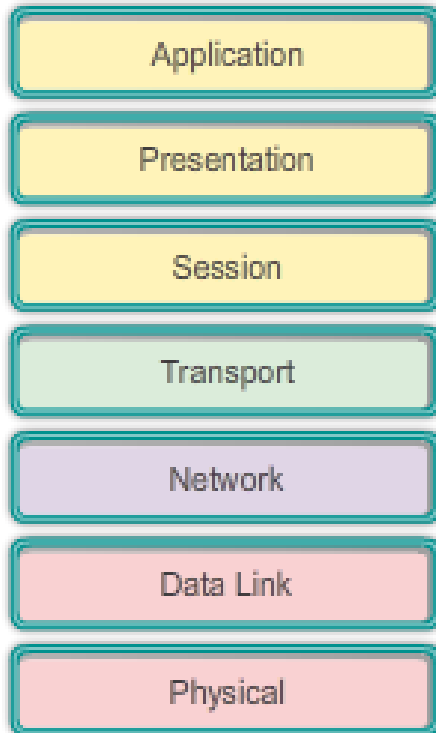
The logical topology and physical media determine the data link protocol used:

- Ethernet
- 802.11 Wireless
- Point-to-Point (PPP)
- High-Level Data Link Control (HDLC)
- Frame-Relay

Each protocol performs media access control for specified logical topologies.



The Frame



- **Header:** Contains control information, such as addressing, and is located at the beginning of the PDU.
- **Data:** Contains the IP header, transport layer header, and application data.
- **Trailer:** Contains control information for error detection added to the end of the PDU.



